Project Kaatru: Launch Brief

City air-quality sensor network · Prepared by the Kaatru programme office · 3 September 2026

> Fictional document created for a live RAG demo. Names, numbers and events are invented.

1. What Project Kaatru is

Project Kaatru ("wind" in Tamil) installs a network of low-cost air-quality sensors across the city of Vellimalai and publishes readings every five minutes on a public dashboard and in the Kaatru mobile app. The goal is to give residents, schools and hospitals street-level pollution data instead of one reading per district.

2. Key facts

Item	Value
Sensors in phase 1	240 sensors across 18 wards
Public launch date	12 November 2026
Phase 1 budget	₹3.8 crore, of which ₹2.1 crore is hardware
Reading interval	Every 5 minutes
Pollutants measured	PM2.5, PM10, NO_2, ozone, temperature and humidity
Programme lead	Dr. Meera Ilango (fictional)
Technology partner	Thendral Systems (fictional)

3. Alert levels

The app sends a push alert when the PM2.5 average over one hour crosses a threshold near the user's saved locations.

Level	PM2.5 (µg/m³, 1-hour average)	What the app says
Green	0 to 30	Good for everyone

Amber	31 to 60	Sensitive groups should limit long outdoor exercise
Red	61 to 120	Everyone should reduce outdoor activity; schools move sports indoors
Purple	above 120	Health advisory issued to hospitals and the city control room

4. Rollout plan

1. **Pilot (September–October 2026):** 40 sensors in wards 3, 7 and 12, near schools and the bus terminus.
2. **Calibration (October 2026):** every sensor is co-located for 14 days with the government reference station and must read within ±10% of it.
3. **Public launch (12 November 2026):** all 240 sensors live, dashboard and app released.
4. **Phase 2 (2027):** expand to 600 sensors and add noise monitoring.

5. Data and privacy

Sensors do not collect any personal data. The app stores saved locations only on the phone. Raw readings are published as open data under a CC BY 4.0 licence and kept for 5 years. Sensors that stop reporting for more than 30 minutes are flagged to the maintenance team, whose target is to repair them within 48 hours.

6. Risks

Risk	Mitigation
Sensor drift over time	Recalibrate against the reference station every 6 months
Vandalism or theft	Mount at 3.5 m height on streetlight poles with tamper alerts
Monsoon power cuts	Each sensor has a 72-hour battery backup
Public confusion about numbers	Colour levels and plain-language advice in Tamil and English